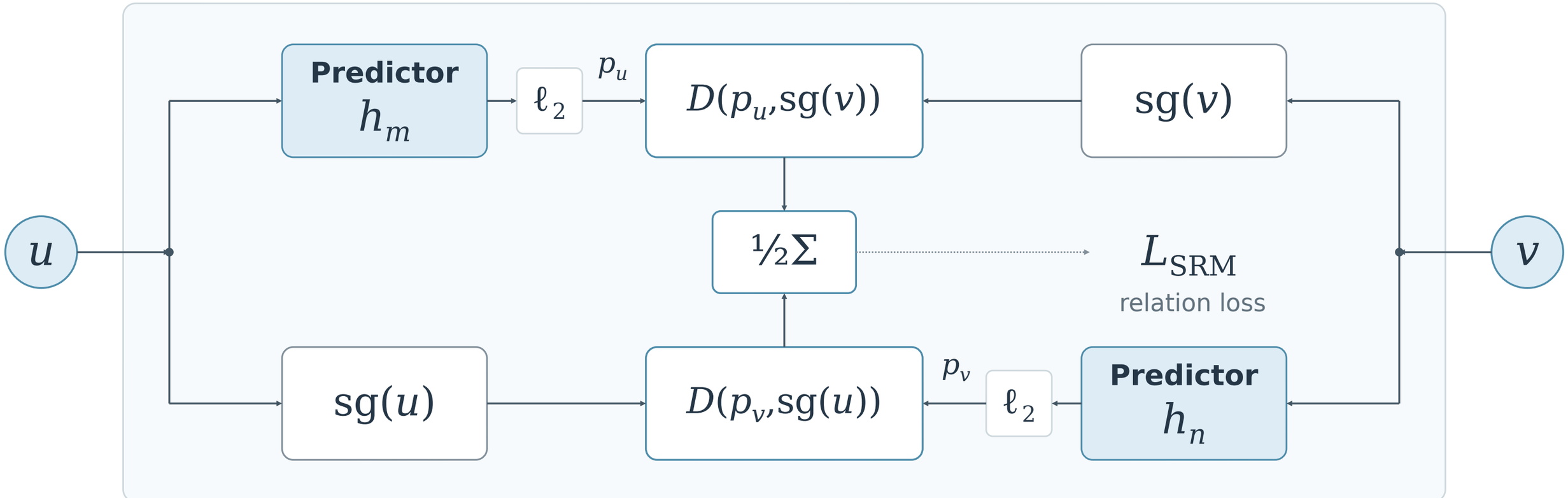
SimSiam Relation Module (SRM)

Symmetric stop-gradient matching of projected 256-D features



$$L_{\mathrm{SRM}}=\tfrac{1}{2}[D(p_u,\mathrm{sg}(v))+D(p_v,\mathrm{sg}(u))]$$

1 · I–I	2 · I–Prompt	3 · I–Answer	4 · Prompt–Answer
$(u,v)=(v_1,v_2)$	$(u,v)=(a^c,t_P)$	$(u,v)=(a^c,t_A)$	$(u,v)=(t_P,t_A)$
$(h_m,h_n)=(h_{\mathrm{I}},h_{\mathrm{I}})$	$(h_m,h_n)=(h_{\mathrm{I}},h_{\mathrm{T}})$	$(h_m,h_n)=(h_{\mathrm{I}},h_{\mathrm{T}})$	$(h_m,h_n)=(h_{\mathrm{T}},h_{\mathrm{T}})$

sg: stop-gradient ℓ_2: L2 normalization D: negative cosine Predictor: 256 → 128 → 256

Predictors are shared by modality across the four calls. PI/PT remain in the shared representation stem.